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How the Yield of Maleic Anhydride in *n*-Butane Oxidation, Using VPO Catalysts, was Improved Over the Years

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Abstract Key patents for the oxidation of *n*-butane to maleic anhydride (MA) using V/P mixed metal oxides (VPO) have been analyzed to evidence the important parameters needed to optimize the catalytic behaviour. The important aspects for the optimization of the catalyst performance are the preparation of the precursor $(VO)HPO_4$-$0.5H_20$, its activation to form the active catalyst $VO_2P_2O_7$ and a small amount of $VOPO_4$, the methodology of the addition of promoters, and the shaping of the catalyst. Even small, sustainable improvements in the MA yield (>1 %) achieved by improved, modified or optimized catalysts are significant on an industrial scale with the world production of MA being a respectable 2.7 Million tons/year. Although substantial improvements in MA yield have been achieved industrially over the past 40 years by improving the VPO catalyst composition and by optimizing process operations; the MA process remains, as practiced today, one of the least efficient industrial selective oxidation processes. Therefore, a huge incentive exists in the field to improve the MA catalyst and process, which led us to search for clues towards this end by analyzing the pertinent patent literature, as reported here.

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1 Introduction

Maleic anhydride is produced commercially from *n*-butane in fixed bed reactors that contain several thousand reaction tubes, either 3.5 or 6 m long, and a diameter of 21–25 mm. The catalysts are variously shaped pellets or rings. In a newer technology practised by Polynt (utilizing Lonza and BP developed technology) fluid bed reactors are employed using fluid bed catalysts. All catalysts are based on the vanadium/phosphorus oxide (VPO) composition.

The highest maleic anhydride yields obtained thus far with these technologies, according to patent literature, are in the range of 57–65 mol%. These are not exceptionally good yields for a selective oxidation process, thus providing much room and opportunity for further improvement.

Several reviews pertaining to the selective oxidation of *n*-butane to maleic anhydride using VPO catalysts exist in the literature [1–4]. The number of references cited in these compilations increased from 62 in the first, to 309 in the forth. Yet in all of these reviews very few references are made to recorded, published patents.

For these reasons it is the purpose of this communication to analyze some of the most relevant patents pertaining to VPO catalyst manufacture, since their preparation and their ultimate efficiency are of great importance to commercial operation in attempts to attain optimum maleic anhydride yields. These factors deemed unimportant and mostly ignored in primary scientific literature are, however of great importance in practical industrial operation and are therefore brought forth in this write-up.

Although statistical methods were employed by the authors of an often cited review article [5] to optimize the parameters of VPO preparations, particularly the V/P ratio, none of the important parameters to industrial operation were considered, since they are very difficult to study on small scale in academic laboratories. Although academic studies are of great importance by shedding light onto the fundamental properties and behaviour of catalysts, they shed little light onto the efficacy needed for optimal commercial operation. In patents it is possible to find important lead aspects that are pertinent to efficient, optimal industrial performance.

Important such aspects for the successful commercial operation of VPO catalysts, found in patent literature comprise the following:

- catalyst life
- pressure drop and density of the catalyst
- environmental and safety problems of catalyst preparation
- time on stream required to attain optimum catalyst performance
- time required for the preparation of the catalyst
- shape of the catalyst
- mechanical strength of the catalyst (particularly important for fluid bed operation)

Important aspects pertaining to commercial catalyst manufacture are:

- preparation of the precursor $(VO)HPO_4 \cdot 0.5H_2O$
- transformation of the precursor to $(VO)_2P_2O_7$ and $VOPO_5$
- equilibration of the optimal V^{4+}/V^{5+} ratio of the active phase under controlled atmosphere
- selection and addition of promoters

Maleic anhydride is produced commercially by either a fixed bed or a fluid bed process using a VPO catalyst and *n*-butane as the feed material.

The fixed bed technology is practiced and licensed by Huntsman (formerly Monsanto), Scientific Design and Technobell. BASF has developed its own fixed bed technology which it practises, but does not license; it licenses however its catalyst. There are two types of fixed bed reactors in commercial use employing: a) 3 m long reaction tubes, or b) 6.5 m reaction tubes. Both employ between 21 and 25 mm diameter tubes. The number of tubes per reactor ranges between 10,000 and 20,000.

The fluid bed technology is newer and is practiced and licensed by Polynt (formerly Lonza) and Ineos (formerly BP). The advantages of fluid bed operation are: a) lower investment cost, b) reduction in downtime (change of catalyst charge), c) greater energy efficiency, and d) larger capacity of a single train. The advantages of fixed bed reactors are: a) higher yields, b) possibility to build larger plants (in parallel), and c) ease of transforming outdated benzene based plants to *n*-butane based plants.

The maleic anhydride yield in industrial reactors is 60 % in fixed bed operation and 55 % in fluid bed operation. The world production is a respectable 2.7 Million tons/year. Nevertheless, is the maleic anhydride process as currently practiced a very inefficient process compared to other commercially practiced selective oxidation processes (e.g., oxidation of propylene to acrylic acid, or ammoxidation of propylene to acrylonitrile). For these reasons there is still ample opportunity to innovate in the field on *n*-butane oxidation to maleic anhydride, and the analysis of patent literature presented here forms a modest basis towards this goal.

2 Discussion

2.1 Prior Art

As "prior art" are considered all of the aspects that have been known in the field of maleic anhydride catalyst manufacture for some time and are now generally accepted and practiced in the field. These include the use of vanadium pentoxide as a source of vanadium and pentavalent or trivalent phosphorus compounds as a source of phosphorous; the use of hydrogen chloride as a reducing agent of V^{5+} to V^{4+} (early on), organic reducing agents (currently), like primary alcohols or secondary aliphatic alcohols such as isobutanol or benzyl alcohol; the preparation of the active catalyst in several stages, with the synthesis of $(VO)HPO_4 \cdot 0.5\ H_2O$ (the precursor), its isolation as a precipitate, drying and shaping and its transformation to $(VO)_2P_2O_7$ (the active phase) by thermal treatment (calcination) and formation as co-products $VO(PO_3)_2$ and/or $VOPO_4$; the performance of VPO mixed oxides can be substantially improved by the addition of promoter elements selected from the groups IA, IB, IIA, IIB, IIIA, IIIB, IVA, IVB, VA, VB, VIA, VIB, and VIIIA of the periodic table of elements, or mixtures thereof.

There is general agreement in the scientific literature on the following points, forming the basis for the need to improve the efficiency of the VPO catalysts, whose answers are now being sought from the patent literature:

1) the best molar yields of maleic anhydride attained thus far lie between 57 and 65 mol% at an *n*-butane conversion of 80–85 mol%;
2) parallel reactions of *n*-butane combustion and oxidative degradation to acetic and acrylic acids impair useful yields

3) side reactions from intermediates formed, also impair useful yields
4) consecutive reactions of maleic anhydride combustion lower its selectivity when the *n*-butane conversion is increased beyond 70 %.

2.2 Synthesis of the Precursor

2.2.1 Standard Oil Company Patent [6]

One of the main innovations in the production of maleic anhydride has been the use of organic reducing agents instead of the highly corrosive HCl to reduce V^{5+} species to form the precursor $(VO)HPO_4$. In a patent issued to Standard Oil Company [6], V_2O_5 was added to isobutanol and refluxed with H_3PO_4 for 16 h to give a slurry that after drying (catalyst precursor) was calcined at 400 °C to form pellets of the active catalyst, which at 380 °C gave a 54.2 % yield of maleic anhydride, while in the reduction with HCl the yield obtained was 43 %.

The role of the organic liquid medium employed was to reduce at least a portion of the V^{5+} to V^{4+}, either upon addition and solvation, or upon mixing and heating. The organic medium should be a solvent for the phosphoric acid but and not for the mixed oxide of vanadium and phosphorus. The use of alcohols as a reducing agent does create some problems for the catalyst, owing to the formation of strongly adsorbed organic residues on its surface and their difficult removal by the subsequent washing and drying treatments. The residues can be physically trapped in between the crystals of the precursor, or physically or chemically trapped ("intercalated") in the crystalline structure of the precursor. It has been found that the amount of organic compounds which remains trapped in the precursor is a fundamental parameter that can adversely affect the performance characteristics of the active catalyst obtained after the thermal treatment.

2.2.2 Lonza Patent [7]

The innovation recorded in a patent issued to Lonza [7] pertains to the preparation of VPO precursors by the reduction with alcohols and the addition of diols. The addition of a diol is an optimal method for reducing the carbon content adsorbed in a vanadium/phosphorus mixed oxide precursor, which when activated leads to superior catalysts for the conversion of *n*-butane to maleic anhydride.The organic medium is comprised of isobutanol or a mixture of isobutanol and benzyl alcohol and/or a polyol. The most preferred polyols are 1,2-ethanediol, 1,3-propanediol, and 1,4-butanediol, preferably containing 5–30 mol% of the polyol with respect to isobutyl alcohol.

In Table 1 are reported the catalytic data of *n*-butane oxidation over VPO catalysts prepared in the presence of isobutanol but absence of a diol, and in the presence of isobutanol and presence of ethanediol or butanediol (diol/isobutanol = 20/80 %).

2.2.3 BASF Patent [8]

In a patent issued to BASF [8] the amount of organic residues on the surface of the precursor after the reduction with alcohols was decreased with a special type of calcination. The catalyst was prepared by reacting vanadium oxides and phosphoric acid in the presence of isobutanol, isolating the formed precipitate, drying it to a content of isobutanol of less than 5 % by weight and thereafter passing over it a gas containing from 0.1 to 9 % oxygen at 190–200 °C. This treatment strongly reduces the amount of organic compounds remaining on the surface of the catalyst and increases the yield and selectivity of maleic anhydride as measured in a monotube of an industrial plant reactor (tube length 6.5 m, tube diameter 22.3 mm) as illustrated in Table 2.

The observed yield differences, although small from the standpoint of an academic laboratory performance are, however, relevant from the standpoint of an industrial reactor, relevant on an industrial scale. Namely, the rule of thumb is that it is cost effective to replace an operating catalyst in an industrial plant with a new improved catalyst when the yield improvement amounts to at least 1 % maleic anhydride. The cost of the new improved catalyst pays off within 1 year, and the catalysts generally last at least 5 years. Thus small yield improvements are very valuable on an industrial scale and are unfortunately often not appreciated or overlooked by catalytic researchers not familiar with industrial operations.

Table 1 Catalyic data from Lonza patent [7]

	Conv. %	Select. %	Yield %
No diol	39.5	66.1	26.1
With ethanediol	47.2	64.5	30.4
With butanediol	75.3	56.8	42.8

Table 2 Effect of oxygen treatment of the precursor on the catalytic behaviour [8]

Catalyst	Temp. °C	S. A. m^2/g	Conv. %	Select. %	Yield %
No treatment	407	27.6	85.5	66.9	57.2
With treatment	401	30.3	85.1	68.2	58.0

2.3 The Formation of $VO_2P_2O_7$

2.3.1 Lonza Patent [9]

In a patent issued to Lonza [9] the innovation realized therein is in the activation of the precursor. The activation of the precursor is performed in a fluid bed reactor, transforming it into the active phase, which is a mixture of vanadyl pyrophosphate $(VO)_2P_2O_7$; vanadyl metaphosphate $VO(P0_3)_2$; and vanadyl phosphate $VOPO_4$. The calcination of the precursor in the prior art has been performed involving the following steps: a) preparation of the precursor; b) transformation of the vanadyl hydrogen phosphate precursor into the active catalyst (mainly vanadyl pyrophosphate) by calcination in air at 350 °C, followed by calcination in nitrogen a 550 °C; c) ageing the catalyst under reaction conditions to activate it to full performance. In the innovation reported in the above patent the activation has been realized inside the fluid bed with the steps reported in Table 3.

The catalyst is fully activated after the steps described in Table 3 have been undertaken. The advantages of this new activation procedure are a better control of the V^{4+}/V^{5+} ratio, higher crystallinity with higher exposure of the 100 crystal face (active face), a fully activated catalyst that does not require aging during the start-up of the catalytic process or make-up, giving steady performance from the start, as soon as it is introduced into the reactor.

2.3.2 Huntsman Patent [10]

In a patent issued to Huntsman [10] the innovation is also related to a special activation of the preformed catalyst. The catalyst's precursor is formed into a shaped body, such as a tablet or pellet, by compression in a die, where a lubricant is generally added to aid in the tabletting or pelleting process. The pellet or tablet is then subjected to calcination to transform the catalyst precursor into an active catalyst. The vanadium–phosphorus catalyst comprises optionally dopants or promoters selected from the group consisting of Zr, Mo, Nb, Cr, Fe, Zn, Ti, V, Mn, Co, Ni, and combinations thereof. The catalyst has a ratio of P/V between 1.03 and 1.10 and a B.E.T. surface area between 25 and 40 m^2/g.

In this invention the active catalyst is formed by exposing a jet shaped, calcined VPO catalyst, having an average vanadium valence between 4.10 and 4.40 to a reduction step with an organic solvent under conditions that facilitate the lowering of the vanadium valence below 4.10. The solvent can be ethylene glycol or propylene glycol or their mixture and the reduction can be carried out below 200 °C. In a subsequent step the solvent is eliminated through drying.

In the reduction of the catalyst with ethylene glycol the average valence of vanadium is reduced from 4.16 to 3.84 and the observed yield of maleic anhydride increases from the original value of 55.5 to 58.3 %. In another example the reduction of the catalyst was performed with ethylene glycol, lowering the vanadium valence from 4.25 to 4.00, resulting in an increase in maleic anhydride yield from 54.1 to 57.2 %.

Table 3 Yield of maleic anhydride using promoted catalysts [11]

Atom %			%	Yield
In	Ta	Sb	SiO_2	%
0	0	0	0	53
0	0	0	2	50
0	5	0	0	27
2	0	0	0	41
5	0	0	2	65
3	0	7	2	72
0	3	2	2	57
0	2	0	2	54
2	3	0	2	59

Fixed bed operation; 380 °C; 1.5 % *n*-butane in the feed

2.4 Methods of Promoter Addition

Several methods of preparation of doped V/P based catalysts are reported in the patent literature. Among these, four methodologies of promoter addition are the preferred ones: a) addition of the promoter to the V^{5+} solution prior to the formation of the precursor, b) addition of the promoter to the preformed precursor, c) addition to the calcined catalyst, and d) addition to the preconditioned catalyst with a flow of *n*-butane and air.

2.4.1 Du Pont Method [11]

In a patent issued to Du Pont de Nemours [11], the catalytic process for the oxidation of *n*-butane to maleic anhydride was improved using a vanadium/phosphorus oxide catalyst containing the promoter silicon and at least one element selected from indium, antimony and tantalum. The Si/V atom ratio was in the range 0.02–3.0:1.0, the (In + Sb + Ta)/V atom ratio in the range 0.005–0.2:1.0, and the P/V atom ratio in the range 0.9–1.3:1.0. This catalyst was prepared in an aqueous or organic liquid medium by the procedure wherein the appropriate vanadium species was substantially reduced to V^{4+} and then contacted with the promoter or promoter precursor and thereafter with the appropriate phosphorus species. While in all other patents the promoters are post-deposited on the VPO catalyst.

From Table 3, where the yields of maleic anhydride are reported, it is apparent that all promoters need the co-presence of silica in order to enhance the maleic anhydride

Table 4 Performance of niobium doped V/P mixed oxides [12]

Type of catalyst	V/Nb	Temp. °C	Conv. %	Select. %	Yield %
No niobium		424	80	59.0	47.20
With niobium	25	379	85	32.8	27.90
With niobium P/V 1. 17	80	374	81	56.0	45.36
With niobium P/V 1.46	80	395	80	66.0	52.80
With niobium P/V 1.17	160	404	81	67.0	54.27

yield, while the presence of silica alone, decreases the maleic anhydride yield.

2.4.2 *Lonza Method [12]*

In a patent issued to Lonza [12] the catalyst was promoted with an amount of niobium corresponding to an atomic ratio of vanadium to niobium in the range of 300:1–80:1, which was added to the preformed precursor. The catalyst exhibits improved activity, improved yield of maleic anhydride, and optimal performance from the very beginning of its catalytic lifetime in relation to prior art preparations. In the prior art, the positive effect of Nb was achieved only when using high amounts of the promoter (low V/Nb atomic ratios) and/or when the thermal treatment of the precursor to transform it into vanadyl pyrophosphate was done inside the reactor, with a mixture of *n*-butane/air, which required long time to reaching high performances.

The activation inside the reactor requires a time period of activation of the catalyst during which the conversion of *n*-butane and the yield of maleic anhydride are far from optimal values, which is very costly and therefore detrimental to commercial applications.

The improvement in catalyst performance using the Lonza invention is achieved by combining the promotion with very low amounts of Nb with a specific method of preparation of the precursor, and a specific thermal treatment to transform it into the active catalyst. The optimal P/V atomic ratio in the preparation mixture is a function of the amount of Nb added. Preferred additional promoter elements are selected from the group consisting of zirconium, bismuth, lithium, molybdenum, boron, zinc, titanium, iron, and nickel.

In Table 4 the temperature of reaction, the conversion of *n*-butane and the molar yield in maleic anhydride for several doped catalysts are reported.

2.4.3 *BASF Method [13]*

In a patent issued to BASF [13] the invention relates to the development of a catalyst for producing maleic anhydride from n-butane which presents as a catalyst a vanadium, phosphorous, mixed oxide doped with iron, with an Fe/V atom ratio preferably from 0.01 to 0.035. Fe(III) phosphate is used as an iron feed stock and the catalyst precursor is shaped before calcination. The catalysts according to this invention have preferably a P/V atom ratio from 1.0 to 1.1, and therefore a slight excess of phosphorous. They have preferably an average oxidation state of vanadium of 4.0–4.3, therefore the presence of some V^{5+} is necessary. The catalysts according to this invention have preferably a BET surface area from 15 to 40 m^2/g, a pore volume from 0.15 to 0.5 ml/g, and a bulk density from 0.5 to 1 kg/l. The iron compound is added onto the dried precursor and therefore the catalyst is doped before calcination to the pyrophosphate.

Table 5 The role of iron as dopant [13]

Catalyst	Surface area m^2/g	Temp. °C	Conv. %	Select. %	Yield %
Undoped	29	419	85.0	67.3	57.3
Doped with Fe	23	402	85.0	69.8	59.5

In table 5 are reported the performances of some of the catalysts tested

The unsupported catalyst may have any form, preferred are cylinders, hollow cylinders, or tri-lobes. Particularly preferred are hollow cylinders with an external diameter from 5 to 7 mm and a height from 3 to 5 mm; and the catalyst may contain additional promoters such as lithium and antimony.

2.4.4 *Ineos Method [14]*

A process for the preparation of a VPO catalyst is described in a patent issued to Ineos [15] where the catalysts is promoted with at least one of the following elements: niobium, cobalt, iron, zinc, molybdenum, or titanium. The catalyst is prepared in three steps:

1) the preparation of a VPO catalyst comprising vanadyl pyrophosphate as the major component and containing less than 5 wt% of vanadyl phosphate,
2) contacting the VPO catalyst with a solution comprising a metal source compound of at least one metal selected from the group consisting of niobium, cobalt, iron, zinc, molybdenum, or titanium with a molar ratio of the metal promoter to vanadium between 0.0015 and 1,
3) drying the metal impregnated catalyst to form the promoted VPO catalyst.

As an alternative to the second step the impregnation with the solution of ammonium niobium nitrate has been carried out on a catalyst that has been previously

Table 6 Catalysts doped with niobium [14]

Type of catalyst	Temp ^{O}C	Conv. %	Select. %	Yield %
Without promoters	428	86.6	65.1	56.4
Niobium added to the precursor	409	86.1	67.9	58.5
Niobium added to calcined catalyst	416	88.7	68.2	60.5
Niobium added to preconditionated catalyst	405	89.8	69.3	62.2

conditioned in *n*-butane/air in order to lower the concentration of $VOPO_4$.

Other promoters can be added to the catalyst, including but not limited to alkali metals, alkaline earth metals, Ti, Cr, W, Ta, U, Co, Mo, Fe, Zn, Hf, Zr, Mn, As, Sb, Te, Bi, Sn, Ge, Nb, Ni, Cu, Cd, Ce, rare earths or mixtures thereof. These promoters may be incorporated into the catalyst precursor in any of the methods known in the art, such as inclusion via the liquid reaction medium prior to or after reduction of the vanadium, or during one or more steps of the preparation of the fluid bed catalyst. The molar ratio of promoter elements to vanadium is 0.001:1–1:1:

In Table 6 the catalytic data of this patent are summarized.

The best performance was obtained with a niobium promoted catalyst, where the promoter was added to the preconditioned catalyst comprised of $(VO)_2P_2O_7$ with a small amount of $VOPO_f$ (<2 %).

2.5 Preparation of Catalysts with Optimal Shape

2.5.1 Monsanto Patent [15]

In a patent issued to Monsanto [16] an improvement in catalytic behaviour of a V/P mixed oxide catalyst was achieved by preparing the catalyst pellet before calcination.

Catalysts for converting *n*-butane to maleic anhydride in high yields at low temperature were prepared by heating a mixture of P_2O_5 with V_2O_5 in HCl solution, forming a putty of the obtained compound, pelletizing the putty, and calcining the pellets to form the catalyst. The optimal P/V ratio was 1.06:1 and the concentration of $V^{4+} > 90$ %. The innovation in the patent is the pelletization prior to calcination.

In prior art catalysts the precursor was calcined before pelletizing, giving a porosity of 30 %, and a yield of maleic anhydride of 29–33 %. In the Monsanto patent it is reported that after the reduction of V^{5+} and addition of phosphorus the excess water was evaporated, the catalyst precursor slurried with 20 % water to form a putty that was subsequently pelletized and calcined 2–4 h at 500 °C. The catalyst prepared according this methodology gave at 500 °C a yield of maleic anhydride of 43 % while a catalyst prepared according to the prior art gave a 20 % maleic anhydride yield. This increase in MA yield is attributed to the preparation method of the catalyst disclosed in this patent. The here disclosed catalyst has a porosity of 53 %, while in prior art catalyst has a porosity of less than 35 %; a substantial difference in porosity, indeed and thereby much better diffusion properties.

2.5.2 Sued Chemie Patent [16]

In a patent issued to Sued Chemie [16] the innovation consists in the optimization of the shaping of the catalyst in order to resolve the problem of the pressure drop across the tubes of the fixed bed reactors employed in maleic acid production. In commercial fixed bed reactors there are several thousand tubes in each reactor, the tubes are long (~10 m) and of small diameter (~2 cm); and they are immersed in a eutectic melt of potassium/sodium–nitrate/nitrite as the heat exchange material. For good catalyst performance it is imperative that there is minimum pressure drop along the reactor tubes.

The Sued Chemie patent addresses this problem whereby the catalyst pellets are prepared from the powder of the precursor containing malonic acid (a pore forming additive) and shaping this precursor mix to hollow cylinders. The shaped catalyst has a cylindrical body with at least one internal hole running right through the body essentially parallel to the cylinder axis of the body with this innovation not only the pressure drop along the reactor tubes has been decreased, but also the yield of maleic anhydride increased.

In prior art practice large catalyst particles were used in order to lower the pressure drop, but these had also lower external surface area and hence also low activity.

The Sued Chemie prepared pellets in the shape of hollow cylinders decrease the density and increase the porosity of the catalytic material thereby increasing the maleic anhydride yield to 58.9 % as compared to 54 % for the old, prior art, catalysts. The density of the newer catalyst was in the range of 0.40–0.50 g/l, the pore volume from 0.38 to 0.50 ml/g.

In a related patent issued to BASF [17] the invention relates to a catalyst shaped body designed as a cylinder having a base surface, a cylinder surface, a cylinder axis and at least one completely penetrating opening running parallel to the cylinder axis; and the base surface of the cylinder has at least four corners The aim is again to decrease the pressure drop along the reactor tubes and to increase the yield of maleic anhydride (from 54.5 to 59.1).

2.5.3 Huntsman Patent [18]

In a patent issued to Huntsman [18] the powder of the catalyst's precursor is shaped into slugs under

compression, converting the precursor to the activated catalyst in three controlled stages. The activated catalyst is granulated and then mixed with a solvent-removable pore building agent in proportions sufficient to provide a pore building agent concentration of between about 6 and 16 % by weight based on the total weight of the activated catalyst. Under compression the activated catalyst can be treated with a solvent removable pore building agent prior to forming it into its predetermined shape. Activating the catalyst precursor prior of tableting increases the yield of maleic anhydride from 58.8 to 61.9 %.

2.6 AgVOx–A New Compound [19]

A new compound, a silver vanadium phosphate has been recently reported by BASF in a patent [19] as a viable catalyst for the conversion of *n*-butane to maleic anhydride.

The invention relates to a silver vanadium phosphate of formula $Ag_aVPbM_cO_d$, where M is at least one element selected from the group consisting of Li, Na, K, Rb, Cs, Mg, Ca, Al, Ga, Si, Nb, Co, Cu, and Zn; and where a is 0.7–1.3; b is 0.8–2.0; c is <0.25; and d is 3.85–8.375 and indicates the number of O^{2-} ions in the formula required to achieve electrical neutrality. The catalysts of the invention can contain one or more additional promoters and their amount is generally not more than about 5 wt%, each calculated as an oxide. Preferably, as additional promoters are compounds of molybdenum, zinc, hafnium, zirconium, titanium, chromium, manganese, nickel, copper, boron, silicon, tin, niobium, cobalt, and bismuth, and in particular molybdenum, zinc, and bismuth. The precursor is prepared by adding phosphorous compounds to a solution containing vanadium and silver species. This catalyst is more active than previous doped catalysts prepared by prior art.

3 Conclusions

An analysis of some relevant patents pertaining to catalysts for the selective oxidation of *n*-butane to maleic anhydride revealed several important factors useful for the future improvement of the catalytic behaviour. Some of these are:

a) Optimization of the precursor's formation by decreasing the amount of organic residues formed after the reduction of V^{5+} species with alcohols on the surface of the precursor with special treatment of the solution containing the precursor.
b) Optimization of the transformation of the precursor to the active phase controlling the amount of V^{5+} formed and in particularly the amount of $VOPO_4$ formed.
c) Addition of different promoters realizing the importance of the different stages of the formation of the active catalyst where the promoters are added.
d) Optimization of the shape and physical properties of the catalyst by choosing different organic compounds (pore expanders) to optimize the porosity.
e) Addition of silver to vanadium and phosphorous, not as a dopant, but to form a stable ternary compound.

The invention of the VPO catalyst by Monsanto [20] and Chevron [21] was a great achievement that displaced the oxidation of benzene to produce maleic anhydride in favour of using *n*-butane as the preferred and less wasteful and less expensive feed material. The commercially attained maleic anhydride yields have been improved from about 35 mol% in the early days of the VPO catalyst invention to 65 mol% today.

Although the invention of the VPO catalyst and its use in the *n*-butane to maleic anhydride selective oxidation process are great accomplishments and significant progress in the yield structure (from 35 to 65 mol%) has been achieved over the past 40 years [1–21], they pale in their efficacy to other commercial selective oxidation catalysts and processes such as the SOHIO fluid bed ammoxidation process where 83 mol% acrylonitrile yield is commercially obtained using propylene, ammonia and air as feeds, or the two stage fixed bed selective oxidation of propylene to acrylic acid where a 90 mol% acrylic acid yield is achieved [22].

With a world production of maleic anhydride of 2.7 Million tons/year, each 1 % improvement in maleic anhydride yield is a welcome saving/profit for industrial operators of this process. It is estimated that a new catalyst with a 1 % improved maleic anhydride yield pays off for itself within 1 year of operation and the catalysts have generally a minimum life time of about 5 years. It becomes a substantial saving.

Considering these facts it becomes apparent that there is still ample room for improvement in the maleic anhydride catalyst innovation and optimization, as well as an optimization of the fixed bed catalyst composition/loading/stacking or optimization of the fluid bed catalyst composition (doping), as well as reactor operation.

Some of the ideas that might lead towards a realization of these goals can be gleaned from the patent literature cited and analyzed here.

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